Q1 - 2024 (04 Apr Shift 1)

Let $\alpha \in (0, \infty)$ and $A = \begin{bmatrix} 1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. If $\det\left(\operatorname{adj}\left(2A - A^T\right) \cdot \operatorname{adj}\left(A - 2A^T\right)\right) = 2^8$, then $(\det(A))^2$ is equal to:

(1) 36

(2) 16

(3) 1

(4) 49

Q2 - 2024 (04 Apr Shift 1)

Let A be a 3×3 matrix of non-negative real elements such that $A\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Then the maximum value of $\det(\mathrm{A})$ is ______

Q3 - 2024 (04 Apr Shift 2)

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = I + \operatorname{adj}(A) + (\operatorname{adj} A)^2 + \ldots + (\operatorname{adj} A)^{10}$. Then, the sum of all the elements of the matrix B is:

(1) -124

(2) 22

(3) -88

(4) -110

Q4 - 2024 (04 Apr Shift 2)

Let A be a 2×2 symmetric matrix such that $A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ and the determinant of A be 1 . If $A^{-1} = \alpha A + \beta I$, where I is an identity matrix of order 2×2, then $\alpha + \beta$ equals ________

Q5 - 2024 (05 Apr Shift 1)

Let A and B be two square matrices of order 3 such that $|A|=3$ and $|B|=2$. Then $\left|\mathrm{A}^{\mathrm{T}} \mathrm{A}(\operatorname{adj}(2 \mathrm{~A}))^{-1}(\operatorname{adj}(4 \mathrm{~B}))(\operatorname{adj}(\mathrm{AB}))^{-1} \mathrm{AA}^{\mathrm{T}}\right|$ is equal to :

(1) 108

(2) 32

(3) 81

(4) 64

Q6 - 2024 (05 Apr Shift 2)

Let $\alpha \beta \neq 0$ and $A=\begin{bmatrix}\beta & \alpha & 3 \\ \alpha & \alpha & \beta \\ -\beta & \alpha & 2 \alpha\end{bmatrix}$. If $B=\begin{bmatrix}3 \alpha & -9 & 3 \alpha \\ -\alpha & 7 & -2 \alpha \\ -2 \alpha & 5 & -2 \beta\end{bmatrix}$ is the matrix of cofactors of the elements of A, then $\operatorname{det}(A B)$ is equal to :

(1) 64

(2) 216

(3) 343

(4) 125

Q7 - 2024 (06 Apr Shift 2)

If A is a square matrix of order 3 such that $\operatorname{det}(A)=3$ and $\operatorname{det}\left(\operatorname{adj}\left(-4 \operatorname{adj}\left(-3 \operatorname{adj}\left(3 \operatorname{adj}\left((2 \mathrm{~A})^{-1}\right)\right)\right)\right)\right)=2^{\mathrm{m}} 3^{\mathrm{n}}$, then $\mathrm{m}+2 \mathrm{n}$ is equal to :

(1) 2

(2) 3

(3) 6

(4) 4

Q8 - 2024 (08 Apr Shift 1)

Let $A=\begin{bmatrix}2 & a & 0\\ 1 & 3 & 1\\ 0 & 5 & b\end{bmatrix}$. If $A^3=4A^2-A-21I$, where I is the identity matrix of order 3×3, then $2a+3b$ is equal to

(1) -9

(2) -13

(3) -10

(4) -12

Q9 - 2024 (08 Apr Shift 1)

Let $A=\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^n, then n is equal to__________

Q10 - 2024 (09 Apr Shift 1)

Let A be a non-singular matrix of order 3 . If $\det(3\operatorname{adj}(2\operatorname{adj}((\det A)A)))=3^{-13}\cdot2^{-10}$ and $\det(3\operatorname{adj}(2\mathrm{~A}))=2^{\mathrm{m}}\cdot3^{\mathrm{n}}$, then $|3\mathrm{~m}+2\mathrm{n}|$ is equal to

Q11 - 2024 (09 Apr Shift 2)

Let $B=\begin{bmatrix}1 & 3\\ 1 & 5\end{bmatrix}$ and A be a 2×2 matrix such that $AB^{-1}=A^{-1}$. If $BCB^{-1}=A$ and $C^4+\alpha C^2+\beta I=O$, then $2\beta-\alpha$ is equal to

(1) 16

(2) 2

(3) 8

(4) 10

Answer Key

Q1 (2)	**Q2** (27)	**Q3** (3)	**Q4** (5)
Q5 (4)	**Q6** (2)	**Q7** (4)	**Q8** (2)
Q9 (7)	**Q10** (14)	**Q11** (4)	

Q1

$$\left|\operatorname{adj}\left(\mathrm{A}-2\,\mathrm{A}^{\mathrm{T}}\right)\left(2\,\mathrm{A}-\mathrm{A}^{\mathrm{T}}\right)\right|=28$$

$$\left|\left(\mathrm{A}-2\,\mathrm{A}^{\mathrm{T}}\right)\left(2\,\mathrm{A}-\mathrm{A}^{\mathrm{T}}\right)\right|=24$$

$$\left|\mathrm{A}-2\,\mathrm{A}^{\mathrm{T}}\right|\left|2\,\mathrm{A}-\mathrm{A}^{\mathrm{T}}\right|=\pm 16$$

$$\left(\mathrm{A}-2\,\mathrm{A}^{\mathrm{T}}\right)^{\mathrm{T}}=\mathrm{A}^{\mathrm{T}}-2\,\mathrm{A}$$

$$\left|\mathrm{A}-2\,\mathrm{A}^{\mathrm{T}}\right|=\left|\mathrm{A}^{\mathrm{T}}-2\,\mathrm{A}\right|$$

$$\Rightarrow\left|\mathrm{A}-2\,\mathrm{A}^{\mathrm{T}}\right|^{2}=16$$

$$\left|\mathrm{A}-2\,\mathrm{A}^{\mathrm{T}}\right|=\pm 4$$

$$\begin{bmatrix}1&2&\alpha\\1&0&1\\0&1&2\end{bmatrix}-\begin{bmatrix}2&2&0\\4&0&2\\2\alpha&2&4\end{bmatrix}$$

$$\begin{vmatrix}-1&0&\alpha\\-3&0&-1\\-2\alpha&-1&-2\end{vmatrix}$$

$$1+3\alpha=4$$

$$3\alpha=3$$

$$\alpha=1$$

$$|\mathrm{A}|=\begin{vmatrix}1&2&1\\1&0&1\\0&1&2\end{vmatrix}=-1-3=-4$$

$$|\mathrm{A}|^{2}=16$$

Q2

$$\text{Let } A=\begin{bmatrix}a_1&a_2&a_3\\b_1&b_2&b_3\\c_1&c_2&c_3\end{bmatrix}$$

$$A\begin{bmatrix}1\\1\\1\end{bmatrix}=3\begin{bmatrix}1\\1\\1\end{bmatrix}$$

$\Rightarrow a_1+a_2+a_3=3$ ...(1)

$\Rightarrow b_1+b_2+b_3=3$ ...(2)

$\Rightarrow c_1+ca_2+c_3=3$ ...(3)

Now,

$|A| = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2)$
$- (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$

$\therefore$ From above in formation, clearly $|A|_{max} = 27$, when $a_1 = 3, b_2 = 3, c_3 = 3$

Q3

$$\text{Adj}(A) = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$(\text{Adj}A)^2 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$$

$$(\text{Adj}A)^{10} = \begin{bmatrix} 1 & -20 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} + \ldots + \begin{bmatrix} 1 & -20 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 11 & -110 \\ 0 & 11 \end{bmatrix} \Rightarrow \text{ sum of elements of B}$$

$= -88$

Q4

Let $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

$\begin{bmatrix} a & b \\ b & d \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}, ad - b^2 = 1$

$a + b = 3, b + d = 7, (3 - b)(7 - b) - b^2 = 1$

$21 - 10b = 1 \to b = 2, a = 1, d = 5$

$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, A^{-1} = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$

$A^{-1} = \alpha A + \beta I$

$\begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} \alpha + \beta & 2\alpha \\ 2\alpha & 5\alpha + \beta \end{bmatrix}$

$\alpha = -1, \beta = 6 \to \alpha + \beta = 5$

Q5

$|A|=3,|B|=2$

$\left|A^T A(\operatorname{adj}(2A))^{-1}(\operatorname{adj}(4B))(\operatorname{adj}(AB))^{-1} A A^T\right|$

$= 3\times 3\times \left|\left(\operatorname{adj}(2A)^{-1}\right| \times |\operatorname{adj}(4B)| \times \left|(\operatorname{adj}(AB))^{-1}\right| \times 3\times 3$

$\downarrow \qquad \downarrow \qquad \downarrow$

$\frac{1}{|\operatorname{adj}(2A)|} \qquad 2^{12}\times 2^2 \qquad \frac{1}{|\operatorname{adj}(AB)|}$

$=\frac{1}{2^6|\operatorname{adj} A|} \qquad =\frac{1}{|\operatorname{adj} B\cdot \operatorname{adj} A|}$

$=\frac{1}{2^6\cdot 3^2} \qquad =\frac{1}{2^2\cdot 3^2}$

$=3^4\cdot\frac{1}{2^6\cdot 3^2}\cdot 2^{12}\cdot 2^2\cdot\frac{1}{2^2\cdot 3^2}=64$

Q6

Equating co-factor fo A_{21}

$\left(2\alpha^2-3\alpha\right)=\alpha$

$\alpha=0,2$ (accept)

Now, $2\alpha^2-\alpha\beta=3\alpha$

$\alpha=2 \quad \beta=1$

$|AB|=|A\operatorname{cof}(A)|=|A|^3$

$$A=\begin{vmatrix}1 & 2 & 3\\ 2 & 2 & 1\\ -1 & 2 & 4\end{vmatrix}=6-2(9)+3(6)=6$$

Q7

$|\mathrm{A}|=3$

$\left|\operatorname{adj}\left(-4 \operatorname{adj}\left(-3 \operatorname{adj}\left(3 \operatorname{adj}\left((2 \mathrm{~A})^{-1}\right)\right)\right)\right|\right.$

$\left|-4 \operatorname{adj}\left(-3 \operatorname{adj}\left(3 \operatorname{adj}(2 \mathrm{~A})^{-1}\right)\right|^2\right.$

$4^6\left|\operatorname{adj}\left(-3 \operatorname{adj}\left(3 \operatorname{adj}(2 \mathrm{~A})^{-1}\right)\right)\right|^2$

$2^{12} \cdot 3^{12}\left|3 \operatorname{adj}(2 \mathrm{~A})^{-1}\right|^8$

$2^{12} \cdot 3^{12} \cdot 3^{24}\left|\operatorname{adj}(2 \mathrm{~A})^{-1}\right|^8$

$2^{12} \cdot 3^{36}\left|(2 \mathrm{~A})^{-1}\right|^{16}$

$2^{12} \cdot 3^{36} \dfrac{1}{|2 \mathrm{~A}|^{16}}$

$2^{12} \cdot 3^{36} \dfrac{1}{2^{48}|\mathrm{~A}|^{16}}$

$2^{12} \cdot 3^{36} \dfrac{1}{2^{48} \cdot 3^{16}}$

$\dfrac{3^{20}}{2^{36}}=2^{-36} \cdot 3^{20}$

$\mathrm{m}=-36 \quad \mathrm{n}=20$

$\mathrm{m}+2 \mathrm{n}=4$

Q8

$A^3-4 A^2+A+21 I=0$

$\operatorname{tr}(A)=4=5+6 \Rightarrow b=-1$

$|A|=-21$

$-16+a=-21 \Rightarrow a=-5$

$2 a+3 b=-13$

Q9

$A=\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}$

$A^2=\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}3 & -3\\ 3 & 0\end{bmatrix}$

$A^3=\begin{bmatrix}3 & -3\\ 3 & 0\end{bmatrix}\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}3 & -6\\ 6 & -3\end{bmatrix}$

$A^4=\begin{bmatrix}3 & -6\\ 6 & -3\end{bmatrix}\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}0 & -9\\ 9 & -9\end{bmatrix}$

$A^5=\begin{bmatrix}0 & -9\\ 9 & -9\end{bmatrix}\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}-9 & -9\\ 9 & -18\end{bmatrix}$

$A^6=\begin{bmatrix}-9 & -9\\ 9 & -18\end{bmatrix}\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}-27 & 0\\ 0 & -27\end{bmatrix}$

$A^7=\begin{bmatrix}-27 & -0\\ 0 & -27\end{bmatrix}\begin{bmatrix}-54 & 27\\ -27 & -27\end{bmatrix}=\begin{bmatrix}3^6\times 2 & -27^2\\ 27^2 & 3^6\end{bmatrix}$

$3^7=3^n\Rightarrow n=7$

Q10

$|3\,\mathrm{adj}(2\,\mathrm{adj}(|A|A))|=|\,3\,\mathrm{adj}\left(2|A|^2\,\mathrm{adj}(A)\right|$

$=|3.2^2|A|^4\,\mathrm{adj}\left(\mathrm{adj}(A)\right|=2^6 3^3|A|^{12}|A|^4$

$=2^6 3^3|A|^{16}=2^{-10}3^{-13}$

$\Rightarrow |A|^{16}=2^{-16}3^{-16}\Rightarrow |A|=2^{-1}3^{-1}$

Now $|3\,\mathrm{adj}(2\,\mathrm{A})|=|3.2^2\,\mathrm{adj}(\mathrm{A})|$

$=2^6 3^3|\mathrm{A}|^2=2^{-\mathrm{m}}3^{-\mathrm{n}}$

$\Rightarrow 2^6 3^3 2^{-2}3^{-2}=2^{-\mathrm{m}}3^{-\mathrm{n}}$

$\Rightarrow 2^{-\mathrm{m}}3^{-\mathrm{n}}=2^4 3^1$

$\Rightarrow \mathrm{m}=-4,\mathrm{n}=-1$

$\Rightarrow |3\,\mathrm{m}+2\mathrm{n}|=|-12-2|=14$

Q11

$\mathrm{BCB}^{-1}=\mathrm{A}$

$\Rightarrow \left(\mathrm{BCB}^{-1}\right)\left(\mathrm{BCB}^{-1}\right)=\mathrm{A}\cdot\mathrm{A}$

$\Rightarrow \mathrm{BCICB}^{-1}=\mathrm{A}^2$

$\Rightarrow \mathrm{BC}^2\,\mathrm{B}^{-1}=\mathrm{A}^2$

$\Rightarrow \mathrm{B}^{-1}\left(\mathrm{BC}^2\,\mathrm{B}^{-1}\right)\mathrm{B}=\mathrm{B}^{-1}\,(\mathrm{A.A})\mathrm{B}$

From equation (1)

$C^2 = A^{-1} \cdot A \cdot B$
$C^2 = B$
Also $AB^{-1} = A^{-1}$
$\Rightarrow AB^{-1} \cdot A = A^{-1} A = I$
$\Rightarrow A^{-1}\left(AB^{-1} A\right) = A^{-1} I$
$B^{-1} A = A^{-1}$
Now characteristics equation of C^2 is
$|C_2 - \lambda I| = 0$
$|B - \lambda I| = 0$
$\Rightarrow \begin{vmatrix} 1-\lambda & 3 \\ 1 & 5-\lambda \end{vmatrix} = 0$
$\Rightarrow (1-\lambda)(5-1) - 3 = 0 \Rightarrow \left(\lambda^2 - 6\lambda + 5\right) - 3 = 0$
$\Rightarrow \lambda^2 - 6\lambda + 2 = 0$
$\Rightarrow \beta^2 - 6B + 2I = 0$
$\Rightarrow C^4 - 6C^2 + 2I = 0$
$\alpha = -6$
$\beta = 2$
$\therefore 2\beta - \alpha = 4 + 6 = 10$